

IIR filter design

Bilinear Transformation

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Bilinear transformation method for IIR filter design

This method overcomes the limitation of the previous method.

Transforms the $j\omega$ axis into unit $\odot$ only once.

Avoids aliasing of frequency components

All points in LHP are mapped inside unit $\odot$
" RHP " outside "

Consider an analog filter with system function

$$H(s) = \frac{b}{s+a} = \frac{Y(s)}{X(s)}$$

$$Y(s)(s+a) = X(s)b$$

This system can also be characterized by the differential equation,

$$\frac{dy}{dt} + ay(t) = bx(t) \quad \text{————— ①}$$

Integrate the derivative and approximate the integral using trapezoidal formula,

$$y(t) = \int_{t_0}^t y'(\tau) d\tau + y(t_0) \quad \text{————— (2)}$$

Approximate this integral by trapezoidal formula, at $t = nT$ and $t_0 = nT - T$

$$\int_{t_0}^t y'(\tau) d\tau = \frac{t - t_0}{2} [y'(t_0) + y'(t)]$$

Substitute for t and t_0 ,

$$\int_{t_0}^t y'(\tau) d\tau = \frac{nT - nT + T}{2} [y'(nT) + y'(nT - T)]$$

Substitute in equation (2),

$$y(nT) = \frac{T}{2} [y'(nT) + y'(nT - T)] + y(nT - T) \quad \text{————— (3)}$$

Evaluate the differential eqn in ① at $t=nT$,

$$\textcircled{1} \Rightarrow \frac{dy(t)}{dt} + ay(t) = bx(t)$$

$$y'(nT) + ay(nT) = bx(nT)$$

$$y'(nT) = -ay(nT) + bx(nT)$$

Substitute this in eqn ③,

$$\textcircled{3} \Rightarrow y(nT) = \frac{T}{2} [y'(nT) + y'(nT-T)] + y(nT-T)$$

$$= \frac{T}{2} [-ay(nT) + bx(nT) - ay(nT-T) + bx(nT-T)] + y(nT-T)$$

Take $y(nT) \equiv y(n)$ and $x(nT) \equiv x(n)$,
and $nT-T = n-1$,

$$y(n) = \frac{T}{2} [-ay(n) + bx(n) - ay(n-1) + bx(n-1)] + y(n-1)$$

Grouping all output & input terms,

$$\left(1 + \frac{aT}{2}\right) y(n) - \left(1 - \frac{aT}{2}\right) y(n-1) = \frac{bT}{2} [x(n) + x(n-1)]$$

Thus we obtain a difference equation

Taking z-transform,

$$\left(1 + \frac{aT}{2}\right) Y(z) - \left(1 - \frac{aT}{2}\right) z^{-1} Y(z) = \frac{bT}{2} [X(z) + z^{-1} X(z)]$$

$$\begin{aligned} H(z) = \frac{Y(z)}{X(z)} &= \frac{\frac{bT}{2} [1 + z^{-1}]}{1 + \frac{aT}{2} - \cancel{z^{-1}} + \frac{aT}{2} z^{-1}} \\ &= \frac{\frac{bT}{2} [1 + z^{-1}]}{(1 - z^{-1}) + \frac{aT}{2} (1 + z^{-1})} \end{aligned}$$

$$\Rightarrow H(z) = \frac{b}{\frac{2}{T} \frac{(1-z^{-1})}{(1+z^{-1})} + \frac{2}{T} \cdot \frac{aT}{2} \frac{(1+z^{-1})}{(1+z^{-1})}}$$

$$= \frac{b}{\frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + a}$$

We know, $H(s) = \frac{b}{s+a}$

Therefore the mapping from s-plane to z-plane is

$$s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$$

This is called bilinear transformation

This transformation holds in general for N^{th} order differential equation too.

Characteristics

$$S = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$$

$$= \frac{2}{T} \left(\frac{z-1}{z+1} \right)$$

$$= \frac{2}{T} \frac{re^{j\omega} - 1}{re^{j\omega} + 1}$$

$$= \frac{2}{T} \left[\frac{r\cos\omega + j r \sin\omega - 1}{r\cos\omega + j r \sin\omega + 1} \right]$$

$$= \frac{2}{T} \left[\frac{r\cos\omega - 1 + j r \sin\omega}{r\cos\omega + 1 + j r \sin\omega} \right] \left[\frac{r\cos\omega + 1 - j r \sin\omega}{r\cos\omega + 1 - j r \sin\omega} \right]$$

$$z = re^{j\omega}$$

$$s = \sigma + j\omega$$

$$e^{j\omega} = \cos\omega + j \sin\omega$$

$$S = \frac{2}{T} \left[\frac{r^2 \cos^2 \omega + r \cos \omega - j r^2 \cos \omega \sin \omega - r \cos \omega - 1 - j r \sin \omega + j r^2 \cos \omega \sin \omega + j r \sin \omega - j^2 r^2 \sin^2 \omega}{(r \cos \omega + 1)^2 - (j r \sin \omega)^2} \right]$$

$$= \frac{2}{T} \times \frac{r^2 + 2j r \sin \omega - 1}{r^2 + 2r \cos \omega + 1}$$

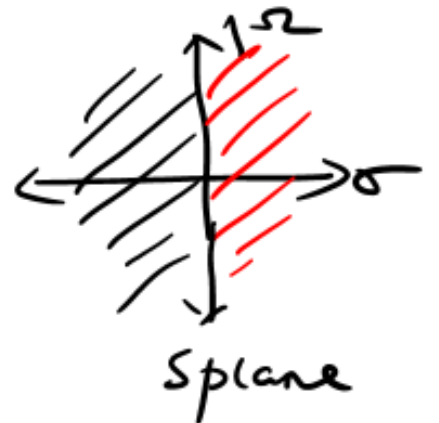
$$S = \left(\frac{2}{T} \left[\frac{r^2 - 1}{r^2 + 2r \cos \omega + 1} + j \frac{2r \sin \omega}{r^2 + 2r \cos \omega + 1} \right] \right)$$

$$S = \sigma + j\omega$$

$$r < 1, \sigma < 0$$

$$r > 1, \sigma > 0$$

$$r = 1, \sigma = 0$$



$$\Omega = \frac{2}{T} \times \frac{\sin \omega}{1 + \cos \omega}$$

$$= \frac{2}{T} \cdot \frac{2 \sin \omega/2 \cos \omega/2}{2 \cos^2 \omega/2}$$

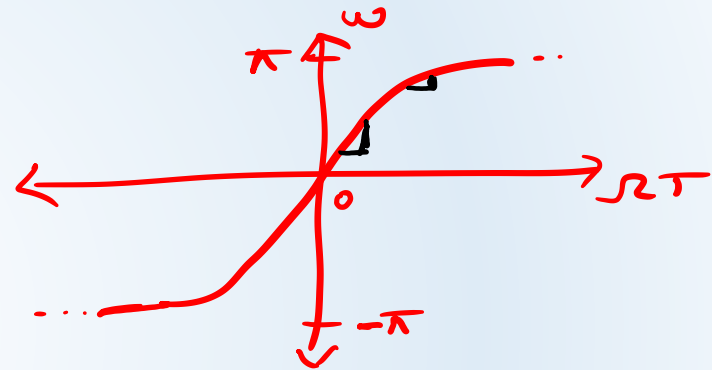
$$\boxed{\Omega = \frac{2}{T} \tan \frac{\omega}{2}}$$

$$\Rightarrow \omega = 2 \tan^{-1} \left(\frac{\Omega T}{2} \right)$$

$\Omega = \pm \infty, \omega = \pm \pi$
one to one mapping

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$



freq compression
warping

nonlinear

Q. Analog filter, $H_a(s) = \frac{s+0.1}{(s+0.1)^2+16}$

Digital filter, using bilinear transform.
 $\downarrow$
 Resonant freq $\omega_r = \pi/2$.

Soln: $H(s) = \frac{s+0.1}{(s+0.1)^2 - (j4)^2} = \frac{s+0.1}{(s+0.1-j4)(s+0.1+j4)}$

Poles, $s_k = -0.1 \pm j4$
 $s = \sigma + j\omega$

$\omega_r = 4$

$\omega_r = \pi/2$

$\omega_r = \frac{2}{T} \tan^{-1} \frac{\omega_r}{2}$

$4 = \frac{2}{T} \tan \frac{\pi/2}{2} \Rightarrow T = 1/2$

$$\begin{aligned}
 H(z) &= H(s) \Big|_{s = \frac{2}{1} \frac{1-z^{-1}}{1+z^{-1}}} = 4 \frac{(1-z^{-1})}{1+z^{-1}} \\
 &= \frac{s+0.1}{(s+0.1)^2 + 16} \Big|_{s = \frac{2}{1} \frac{1-z^{-1}}{1+z^{-1}}} \\
 &= \frac{4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.1}{\left(4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.1 \right)^2 + 16}
 \end{aligned}$$

$$\begin{aligned}
 H(z) &= \frac{0.128 + 0.006 z^{-1} - 0.122 z^{-2}}{1 + 0.995 z^{-2}} \\
 &= 0.128 \frac{(z^2 + 0.046 z - 0.953)}{z^2 + 0.995}
 \end{aligned}$$

Poles @ $0.987 e^{\pm j\pi/2}$
 Zeros @ $-1, 0.953$

3) Matched z-transform

Direct mapping of poles & zeros

$$H(s) = \frac{\prod_{k=1}^M (s - z_k)}{\prod_{k=1}^N (s - p_k)} \xrightarrow{\text{map}} H(z) = \frac{\prod_{k=1}^M (1 - e^{z_k T} z^{-1})}{\prod_{k=1}^N (1 - e^{p_k T} z^{-1})}$$

*Thank
you*

